

Testing Scientific Expectations under ERGMs with BFpack

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How to use this document. Every code block below is ready to paste into an R session and run in order. To keep the handout light, the chunks are *not* executed while knitting (`eval = FALSE` in the setup chunk); instead, the expected results are shown and discussed in the text and tables. To run the analysis live, either change `eval = FALSE` to `eval = TRUE` in the setup chunk, or simply copy each block into R. Fitting the Bayesian ERGMs with `bergm` takes a few minutes per model.

This vignette is based on Mulder, J., Friel, N., & Leifeld, P. (2024), “Bayesian testing of scientific expectations under exponential random graph models,” Social Networks, 78, 40–53 (<https://doi.org/10.1016/j.socnet.2023.11.004>). It adapts the analyses and results reported in ‘Application A’ into a hands-on tutorial for the BFpack package.

Introduction and learning objectives

Exponential random graph models (ERGMs) are the workhorse for explaining how ties form in social and political networks. The standard workflow fits an ERGM by maximum likelihood and then reads off p -values to decide which predictors “matter”. This tutorial teaches a complementary, and in several respects more informative, approach: **Bayesian hypothesis testing with Bayes factors and posterior probabilities**, as implemented in the R package **BFpack**.

The methodology, and the running example we use here, come from:

Mulder, J., Friel, N., & Leifeld, P. (2024). *Bayesian testing of scientific expectations under exponential random graph models*. **Social Networks**, 78, 40–53. <https://doi.org/10.1016/j.socnet.2023.11.004>

The empirical case is the German toxic-chemicals policy network of Leifeld & Schneider (2012), distributed as the `chemnet` data in the `btergm` package.

Why go beyond the p -value?

The paper motivates the Bayesian approach through three concrete limitations of significance testing that bite hard in network research:

1. **A p -value cannot support a null.** It can only *falsify* a hypothesis. If preference similarity is non-significant after controlling for opportunity structures, we cannot tell whether that is *evidence of absence* (the effect really is ~ 0) or merely *absence of evidence* (the study is underpowered). Leifeld & Schneider (2012) ran into exactly this: they wanted to argue that preference similarity had *no* additional effect, but a non-significant p -value cannot license that claim.
2. **The p -value is inconsistent under the null.** Because it is uniform under the null, there is always a fixed probability (typically .05) of rejecting a true null — even as the network grows arbitrarily large. A Bayes factor, by contrast, is *consistent*: evidence for the true hypothesis grows without bound as the network grows.

3. The *p*-value cannot directly test competing constrained hypotheses. Scientific expectations are often phrased with order constraints — “effect A is larger than effect B, which is larger than effect C”. Testing $\beta_{\text{committee}} > \beta_{\text{influence}} > \beta_{\text{pref.sim}} > 0$ against equality or against “none of the above” is natural for a Bayes factor, but awkward or impossible with *p*-values.

Bayes factors address all three: they quantify evidence **for** a null, they are **consistent**, and they test **equality and/or order constraints** on ERGM coefficients directly.

What you will be able to do after this session

By the end you will be able to:

- fit an ERGM with `ergm` and inspect the exact coefficient names with `get_estimates()`;
- translate a substantive expectation into a `BFpack` hypothesis string using equality (=) and order (>, <) constraints;
- compute Bayes factors and posterior probabilities with `BF()` for
 - (i) a precise null hypothesis and (ii) a set of competing constrained hypotheses;
- read the **specification table** to understand how a Bayes factor balances *fit* against *complexity* (an Occam’s razor); and
- report and interpret the results the way the paper does.

Setup: packages, seed, and data

```
# Install once, if needed:
# install.packages(c("statnet", "ergm", "BFpack", "btergm", "sna", "Bergm"))

library("statnet")    # umbrella package: loads network, sna, ergm, ...
library("ergm")       # fitting ERGMs by MLE
library("BFpack")     # Bayes factors for constrained hypotheses (v1.2.3+)
library("btergm")     # ships the chemnet policy-network data
library("sna")        # network descriptives (e.g. betweenness)

seed <- 1234
set.seed(seed)
```

We use a fixed `seed` throughout. Two sources of randomness matter here: the MLE search inside `ergm`, and — more importantly — the MCMC sampler inside the Bayesian ERGM fit that `BFpack` runs under the hood. Fixing the seed makes the workshop reproducible, but note that Bayes-factor values will still wobble a little from run to run; the *substantive* conclusions are stable.

```
data("chemnet")      # ?chemnet for documentation (Leifeld & Schneider 2012, AJPS)
```

The chemnet policy network

`chemnet` concerns political information exchange among 30 organizations (interest groups, government agencies, scientific bodies, ...) involved in German toxic-chemicals policy. Loading it makes several objects available:

Object	Meaning
<code>pol</code>	political/strategic information exchange (the outcome network)
<code>scito</code> , <code>scifrom</code>	reported sending / receiving of scientific information

Object	Meaning
intpos	matrix of actors' positions on policy issues (preferences)
committee	actor-by-committee membership matrix
infrep	influence attribution (who is named as influential)
types	organization type for each actor

Our analysis explains the **political information network pol** using covariates built from the other objects, following the paper.

Preparing the network data

The covariates below reproduce Equations (1)–(4) of the paper. Each transforms raw data into a dyadic predictor that can enter the ERGM through `edgecov()`.

```
# (1) Confirmed scientific-information tie: i->j only if i claims sending
#     AND j claims receiving. Element-wise product of scito and t(scifrom).
sci <- scito * t(scifrom) # Eq. (1)

# (2)-(3) Preference similarity from issue positions: Euclidean distance,
#         then reversed so that LARGER = MORE similar.
prefsim <- dist(intpos, method = "euclidean") # Eq. (2)
prefsim <- max(prefsim) - prefsim # Eq. (3): distance -> similarity
prefsim <- as.matrix(prefsim)

# Standardize preference similarity (needed for the ORDER test, so that
# effect sizes are comparable on a common scale). We standardize the
# off-diagonal entries jointly and write them back.
prefsim_scaled <- c(scale(c(prefsim[lower.tri(prefsim)],
                           prefsim[upper.tri(prefsim)])))

prefsim_st <- prefsim
prefsim_st[lower.tri(prefsim_st)] <- prefsim_scaled[1:(length(prefsim_scaled)/2)]
prefsim_st[upper.tri(prefsim_st)] <- prefsim_scaled[(length(prefsim_scaled)/2 + 1):length(prefsim_scaled)]

# (4) Shared committee memberships: co-membership count, diagonal set to 0.
committee <- committee %*% t(committee) # Eq. (4)
diag(committee) <- 0 # self-membership is meaningless
committee_scaled <- c(scale(c(committee[lower.tri(committee)],
                              committee[upper.tri(committee)])))

committee_st <- committee
committee_st[lower.tri(committee_st)] <- committee_scaled[1:(length(committee_scaled)/2)]
committee_st[upper.tri(committee_st)] <- committee_scaled[(length(committee_scaled)/2 + 1):length(committee_scaled)]

# Organization type as a vertex attribute (vector form).
types <- types[, 1]

# Influence attribution, standardized.
infrep_st <- matrix(c(scale(c(infrep))), nrow = nrow(infrep))
```

Why standardize? For a test that only asks “is this effect zero?” the scale of a covariate is irrelevant. But for an **order-constrained** test such as `committee > influence > pref.sim`,

the coefficients are only comparable if the predictors share a common scale. Standardizing the three edge covariates puts them on the same footing, so the ordering of coefficients is an ordering of *effect sizes*.

Build the network objects

```
# Outcome: political / strategic information exchange
nw.pol <- network(pol)
set.vertex.attribute(nw.pol, "orgtype", types)
set.vertex.attribute(nw.pol, "betweenness", betweenness(nw.pol))

# Covariate network: confirmed technical / scientific information exchange
nw.sci <- network(sci)
set.vertex.attribute(nw.sci, "orgtype", types)
set.vertex.attribute(nw.sci, "betweenness", betweenness(nw.sci))
```

A simplified ERGM to learn the workflow

We first fit a compact six-term ERGM. It is not the final published model, but it is the fastest way to learn the four-step BFpack workflow.

Step 1 — Fit the model

```
model1 <- ergm(
  nw.pol ~ edges +
    mutual + # reciprocity
    edgecov(nw.sci) + # scientific information exchange
    edgecov(prefsim_st) + # preference similarity (standardized)
    edgecov(committee_st) + # shared committees (standardized)
    edgecov(infrep_st), # influence attribution (standardized)
  control = control.ergm(seed = seed)
)
summary(model1)
```

Step 2 — Read the exact parameter names

This step is essential. The names BFpack expects in a hypothesis string are exactly the names printed by `get_estimates()` — not the covariate object names. An `edgecov(prefsim_st)` term becomes the parameter `edgecov.prefsim_st`.

```
get_estimates(model1)
#> Coefficient names include, among others:
#> edges, mutual, edgecov.nw.sci,
#> edgecov.prefsim_st, edgecov.committee_st, edgecov.infrep_st
```

Common pitfall (fixed here). An earlier draft of this script wrote the hypothesis as "edgecov.prefsim = 0". That name does **not** exist — the term is `edgecov.prefsim_st`, so the test would error or silently mismatch. Always copy names verbatim from `get_estimates()`.

Step 3 & 4 — Test a precise null: does preference similarity matter?

The central substantive question from Leifeld & Schneider (2012): once opportunity structures are controlled for, does preference similarity still drive information exchange, or is its effect fully “absorbed”?

```
# Step 3: formulate the hypothesis (note the correct parameter name).
hypo1 <- "edgecov.prefsim_st = 0"

# Step 4: BF() fits the Bayesian ERGM internally (this is the slow part),
# then computes the Bayes factor and posterior probabilities. By default the
# null is tested against its complement (the two-sided alternative).
BF_model1_test1 <- BF(model1, hypothesis = hypo1, main.iters = 2000)
print(BF_model1_test1)
```

`print()` returns the posterior probabilities of the constrained hypothesis H1: $\beta = 0$ and its complement H2: $\beta \neq 0$. The key thing to notice — impossible with a p -value — is that we can obtain **positive evidence in favor of the null**.

```
# Posterior probabilities that EACH coefficient is zero, negative, or positive.
# BFpack produces this exploratory test for every parameter by default.
summary(BF_model1_test1)
```

Interpretation. Suppose the Bayes factor favors the null, $BF_{01} \approx 5$. Read this as: *the observed network is about five times more likely under a model where preference similarity has no effect than under a model where it can take any value*. With equal prior odds this maps to a posterior probability of roughly 0.83 for the null. Compare that to the p -value of .136 from the same data: the p -value only says “not significant”, whereas the Bayes factor says “there is positive evidence the effect is zero — the non-significance is not just low power.” (The exact figures under the full published model are reproduced in the section “The full model, reproducing the paper” below.)

The illustrative numbers above are for orientation; because `model1` is the simplified specification and because the Bayes factor rests on MCMC draws, your exact values will differ slightly. The published values appear under the full model below.

Testing competing constrained hypotheses

Now the feature that has no clean p -value analogue: comparing several substantive theories at once, including **order constraints**.

Based on Leifeld & Schneider we can theorize a *ranking* of importance — shared committees matter most, then influence attribution, then preference similarity. We encode three rival accounts:

- **H1 (ordered):** `committee > influence > pref.sim > 0` — the theorized ranking, all positive.
- **H2 (equal):** `committee = influence = pref.sim > 0` — the three matter equally and positively.
- **H3 (null):** `committee = influence = pref.sim = 0` — none of them matters.

By default `BF()` also adds the **complement** (none of H1–H3 holds), so we in fact compare four hypotheses.

```
# Steps 1 and 2 are unchanged (same fitted model, same parameter names).

# Step 3: formulate the three hypotheses, separated by semicolons.
hypo2 <- "edgecov.committee_st > edgecov.infrep_st > edgecov.prefsim_st > 0;
         edgecov.committee_st = edgecov.infrep_st = edgecov.prefsim_st > 0;
         edgecov.committee_st = edgecov.infrep_st = edgecov.prefsim_st = 0"

# Step 4: compute BFs and posterior probabilities. The complement is added
```

```
# automatically, so the `complement` argument can be omitted.
# NOTE: we test under model1 (which contains all three terms). An earlier
# draft mistakenly passed `model2` here before it was even fitted.
BF_model1_test2 <- BF(model1, hypothesis = hyp2, main.iters = 2000)
print(BF_model1_test2)

# The full output, including the Specification Table, is worth studying:
summary(BF_model1_test2)
```

Reading the specification table

`summary()` prints a **specification table** that decomposes each Bayes factor into two ingredients:

- **relative fit** — how well the constrained hypothesis matches the posterior (posterior density at an equality value; posterior probability that order constraints hold); and
- **relative complexity** — the same quantities under the *prior*.

The Bayes factor of a hypothesis against the unconstrained model is `fit / complexity`. This is the **Occam's razor**: a hypothesis is rewarded for fitting well but penalized for being complex (for carving out a large slice of the parameter space a priori). For a pure order constraint on three parameters, the prior probability of any one ordering is 1/6, so a perfectly supported ordering earns a Bayes factor up to 6 against the unconstrained model.

Changing the prior probabilities of hypotheses

Equal prior probabilities are the default. If theory or prior literature makes some hypotheses more plausible a priori, supply `prior.hyp.conf` (one weight per hypothesis, in order, including the complement):

```
# Example: down-weight H1/H2/H3 and put more prior mass on the complement.
BF_model1_test3 <- BF(model1, hypothesis = hyp2,
  main.iters = 2000,
  prior.hyp.conf = c(1, 1, 1, 4))
print(BF_model1_test3)
```

The Bayes factors (evidence *in the data*) are unchanged; only the posterior probabilities, which combine evidence with prior odds, shift.

The full model, reproducing the paper

We now fit the full specification — Model 2 of Leifeld & Schneider (2012) from the paper. It adds actor-type mixing terms and the geometrically weighted shared-partner statistics (GWESP, GWDSP).

```
model2 <- ergm(
  nw.pol ~ edges +
    edgecov(prefsim_st) +
    mutual +
    nodemix("orgtype", base = -7) +
    nodeifactor("orgtype", base = -1) +
    nodeofactor("orgtype", base = -5) +
    edgecov(committee_st) +
    edgecov(nw.sci) +
    edgecov(infrep_st) +
    gwesp(0.1, fixed = TRUE) +
    gwdsp(0.1, fixed = TRUE),
```

```

control = control.ergm(seed = seed)
)
summary(model2)

```

Under this model the paper runs the same two tests we practiced above:

```

# Test 1: is the effect of preference similarity zero?
hypo1 <- "edgecov.prefsim_st = 0"
BF_model2_test1 <- BF(model2, hypothesis = hypo1, main.iters = 10000)
print(BF_model2_test1)
summary(BF_model2_test1)

# Test 2: the ranking of opportunity-structure effects.
hypo2 <- "edgecov.committee_st > edgecov.infrep_st > edgecov.prefsim_st > 0;
         edgecov.committee_st = edgecov.infrep_st = edgecov.prefsim_st > 0;
         edgecov.committee_st = edgecov.infrep_st = edgecov.prefsim_st = 0"
BF_model2_test2 <- BF(model2, hypothesis = hypo2, main.iters = 10000)
print(BF_model2_test2)
summary(BF_model2_test2)

```

Note on main.iters. For a polished analysis the paper uses `main.iters = 10000` posterior draws for accurate Bayes factors. That is slow. For live exploration in the workshop, keep `main.iters = 2000` (or the `bergm` default) to avoid long waits, then increase it for final results.

Published results

Test 1 — no effect of preference similarity

Under Model 2 the posterior and prior density of $\beta_{\text{pref.sim}}$ at zero are **2.04** and **0.387**, giving

$$BF_{01} = \frac{2.04}{0.387} \approx 5.2,$$

i.e. **positive evidence that preference similarity has no additional effect** on information exchange once opportunity structures are controlled. With equal prior probabilities the posterior probabilities are

$$P(H_1 | Y) = 0.839, \quad P(H_2 | Y) = 0.161.$$

The classical two-sided p -value on the same coefficient is **.136** — “non-significant”, but silent on whether that reflects a true zero or an underpowered study. The Bayes factor resolves the ambiguity in favor of a genuine null. (For reference, the BIC-based evidence is 13.63 and the AIC-based evidence 1.26, bracketing the Bayes factor, since BIC leans toward the simpler model and AIC toward the larger one.)

Table 1. Full coefficient-level results under Model 2 (Application A). Classical estimates, Bayesian estimates under the unit-information prior, and posterior probabilities that each coefficient is zero, negative, or positive (equal prior probabilities).

Term	MLE	s.e.	p	post. mean	post. sd	$P(\beta=0 Y)$	$P(\beta<0 Y)$	$P(\beta>0 Y)$
edges	-4.039	1.290	0.002	-2.525	0.675	—	—	—
pref. sim. (st.)	0.118	0.079	0.136	0.116	0.106	0.722	0.037	0.241
reciprocity	0.808	0.248	0.001	0.765	0.298	0.138	0.005	0.857

Term	MLE	s.e.	p	post. mean	post. sd	$P(\beta=0 \mid Y)$	$P(\beta<0 \mid Y)$	$P(\beta>0 \mid Y)$
int. group ho- mophily	1.067	0.291	0.000	1.222	0.485	0.178	0.005	0.817
gov. target	0.597	0.189	0.002	0.561	0.248	0.259	0.009	0.732
sci. source	0.072	0.218	0.742	0.063	0.269	0.822	0.072	0.106
common com- mittees (st.)	0.727	0.122	0.000	0.839	0.173	0.000	0.000	1.000
sci. com- muni- cation	2.910	0.630	0.000	2.935	1.006	0.039	0.002	0.958
infl. attr. (st.)	0.439	0.088	0.000	0.438	0.114	0.003	0.000	0.997
GWESP(0.1)	2.552	1.129	0.024	1.363	0.597	0.094	0.012	0.895
GWDSP(0.1)	-0.134	0.049	0.007	-0.208	0.087	0.146	0.847	0.007

(No Bayesian test is reported for *edges*: an improper flat prior is used for this nuisance intercept, and flat priors cannot be used for Bayes-factor testing.)

Test 2 — the ranking of opportunity-structure effects

Table 2. Bayes factors and posterior probabilities for the three constrained hypotheses (equal prior probabilities).

Hypothesis	vs H1	vs H2	vs H3	$P(H_t \mid Y)$
H1: committee > influence > pref.sim	1.000	74.335	97.953	.977
H2: committee = influence = pref.sim > 0	0.013	1.000	1.318	.013
H3: committee = influence = pref.sim = 0	0.010	0.759	1.000	.010

The ordered hypothesis H1 receives about **74×** the evidence of the equality hypothesis and about **98×** that of the complement, and carries a posterior probability of roughly **98%**. After seeing the data, then, there is strong support for the theorized ranking: the (standardized) effect of shared committees is largest, followed by influence attribution, followed by preference similarity — consistent with the point estimates in Table 1.

Reporting and interpretation checklist

When you write up a BFpack ERGM analysis, report:

- the **fitted ERGM specification** and how each covariate was constructed;
- the exact **hypotheses** in constraint notation, with their substantive meaning;
- the **Bayes factors** between hypotheses (and/or against the unconstrained model) and the **posterior probabilities**, stating the prior probabilities used;
- for a contested null, the **posterior/prior density ratio at zero** that produces BF_{01} , which mirrors Bayesian estimation; and
- the number of posterior draws (`main.itors`) and the seed, for reproducibility.

Finally note that posterior probabilities live on a **continuous** scale — no dichotomous accept/reject decision is required — and if you *do* commit to a hypothesis, $1 - P(H | Y)$ is the conditional probability of being wrong given this network (e.g. ~16% for the null in Test 1, ~2% for H1 in Test 2).

References

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